

BCB430

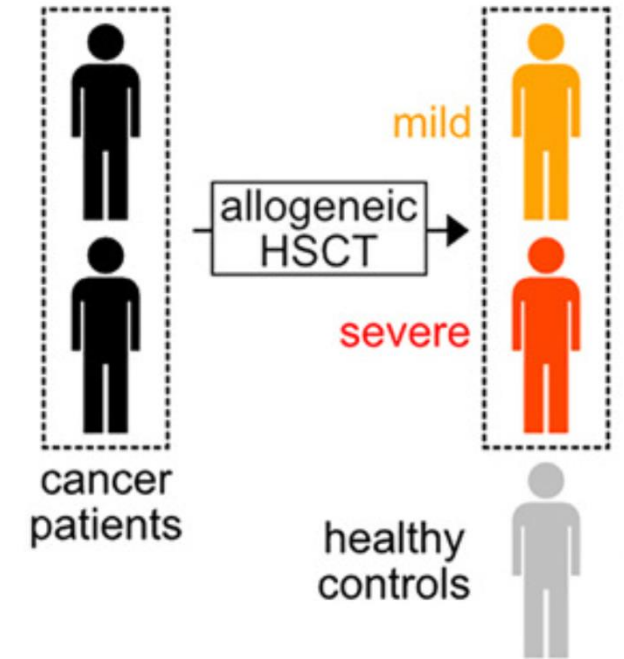
Assignment 3 - Presentation of Pathway and Network Analysis Results

Yuxi Zhang

March 31st 2026

INTRODUCTION- GSE234357

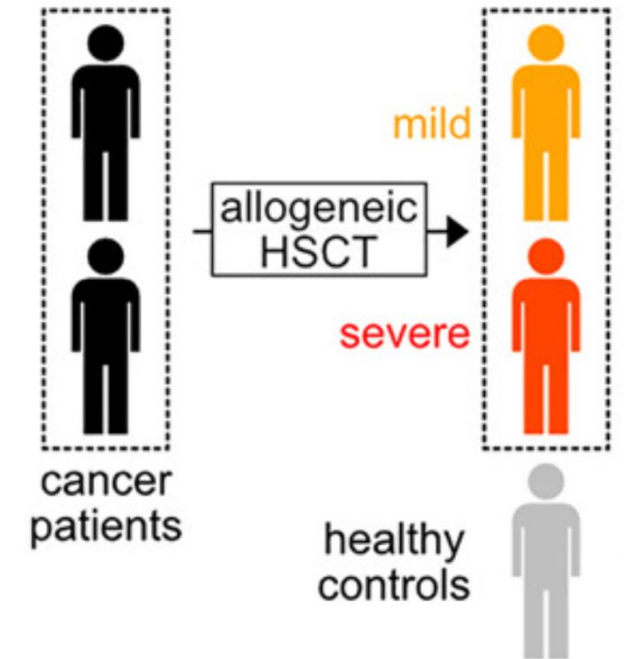
Investigation: transcriptomic and immune landscape of acute graft-versus-host disease (aGvHD) following allogeneic hematopoietic stem cell transplantation (HSCT) .



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Question: Identifying differentially expressed genes and coordinated biological pathways to better understand how the immune landscape and microbiome influence disease severity in post-transplant patients



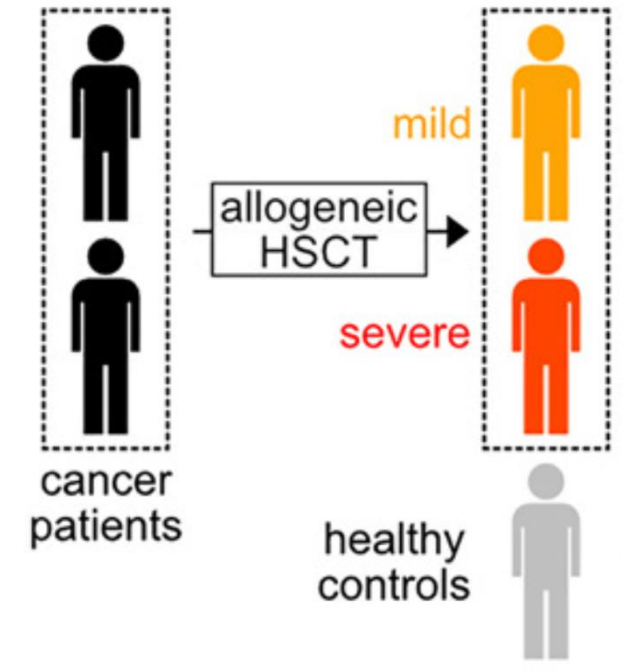
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- **Test:** 6 colon biopsies from HSCT patients with aGvHD
- **Control:** 3 colon biopsies from HSCT patients without aGvHD
- **Data:** multimodal single-cell RNA sequencing (scRNA-seq)

(Jarosch et al. 2023)



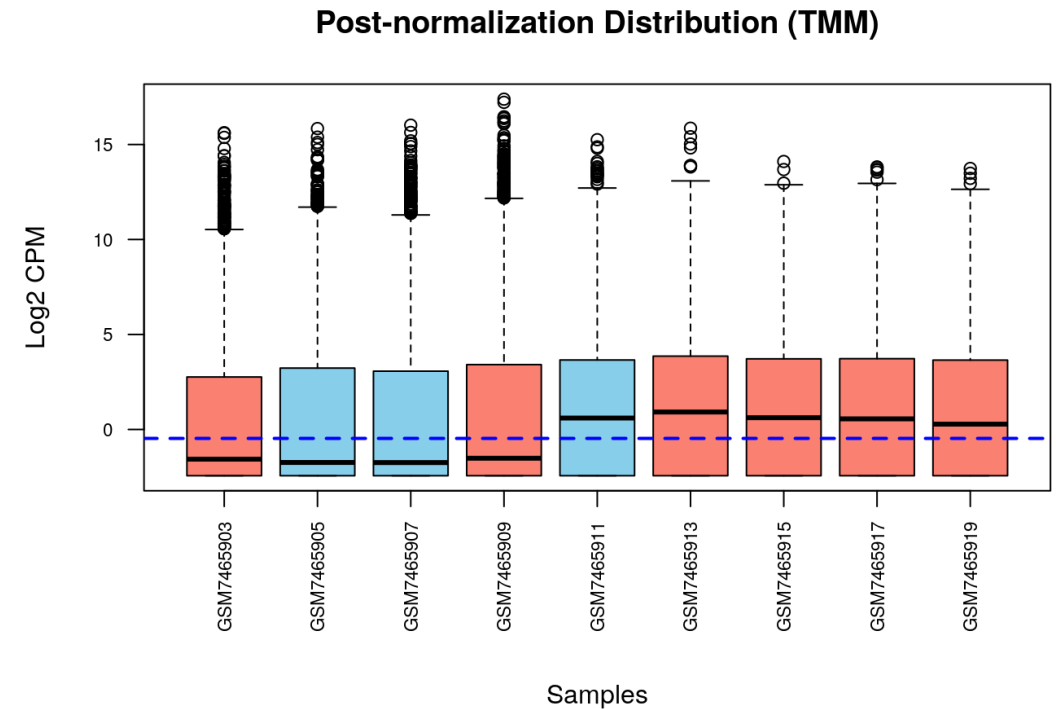
INTRODUCTION – NORMALIZATION/SCORING

Normalization: TMM (Trimmed Mean of M-values)

edgeR package's `calcNormFactors()` function

Rationale:

- Composition bias of RNA between samples
- Handling 25-fold differences in library size
- Ensuring comparability across all replicates

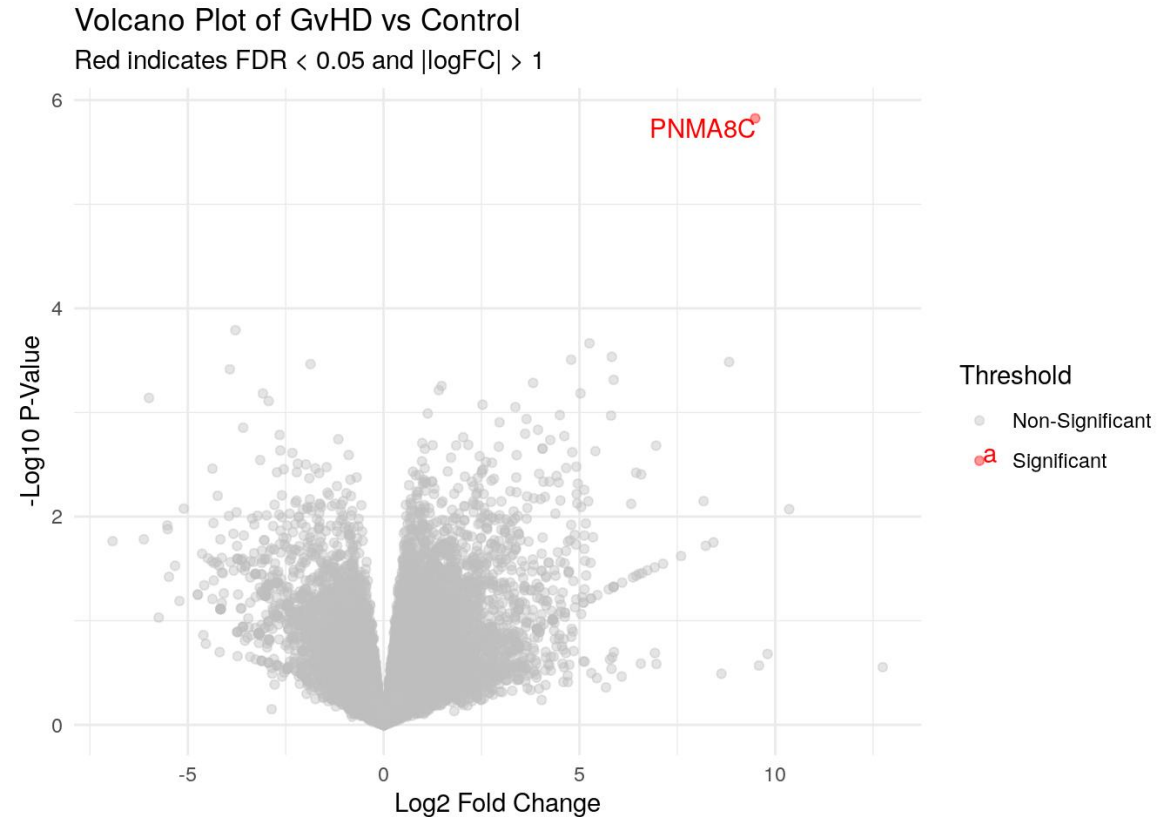


INTRODUCTION – NORMALIZATION/SCORING

Following TMM normalization:

P-values were calculated using the **Quasi-Likelihood (QL) F-test** in edgeR

Thresholds: $\text{FDR} < 0.05$ and $|\log\text{FC}| > 1$



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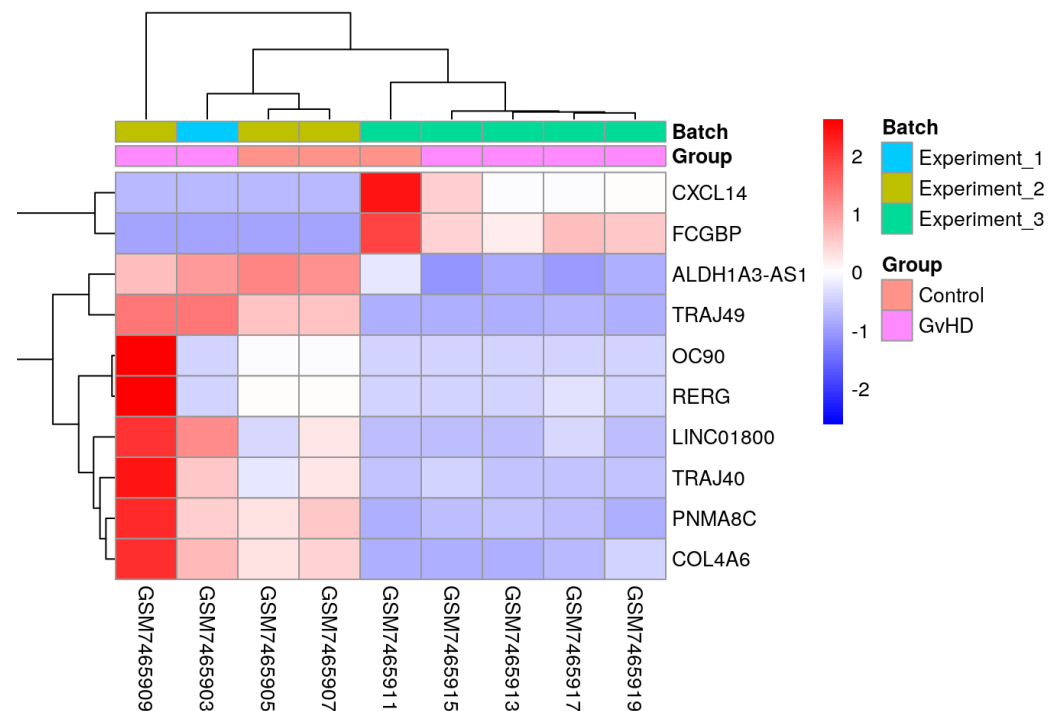
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Visualization Z-score via `scale = "row"`

- Centered each gene's expression to
- Clearly see pattern of up-regulation (GvHD) vs. down-regulation (Control)

Hierarchical Clustering and Heatmap of Top Differentially Expressed Genes





METHOD - NON-THRESHOLDED ENRICHMENT ANALYSIS

Approach: Gene Set Enrichment Analysis (GSEA)

$$Score = sign(logFC) \times -\log_{10}(p\text{-value})$$

Gene set: Bader Lab database

Version: Human_GOBP_AllPathways_no_GO_iea (Ensembl 113 / Genenames 2024)

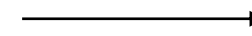
- This GMT file aggregates Gene Ontology: Biological Processes, Reactome, and WikiPathways.

METHOD - NON-THRESHOLDED ENRICHMENT ANALYSIS

Approach: Gene Set Enrichment Analysis (GSEA)

```
gsea_jar, "GseaPreranked",  
"-rnk", rnk_file,  
"-gmx", dest_gmt_file,  
"-collapse false",  
"-nperm 1000",  
"-scoring_scheme weighted",  
"-set_min 15",  
"-set_max 200",
```

24,492 genes



filtering

6,513 genes

RESULTS - NON-THRESHOLDED ENRICHMENT ANALYSIS

Positively enriched pathways:

- a) Translocation of ZAP-70 to Immunological Synapse
- b) Positive Regulation of T Cell Migration
- c) Complement and Coagulation Cascades
- d) Lipid Metabolism

Negatively enriched pathways:

- a) Regulation of T Cell Differentiation in Thymus
- b) IFN-Alpha
- c) Mitochondria Pathway

	na_pos	na_neg
Total out of	4,760	1753
FDR < 25%	0	84
p-value < 1%	39	108
p-value < 5%	193	190

Annotated Enrichment Map of Biological Processes in Graft-versus-Host Disease

RESULTS – CYTOSCAPE

FDR q-value cutoff: 0.25

p-value cutoff: 0.05

NES (GSEA only): All

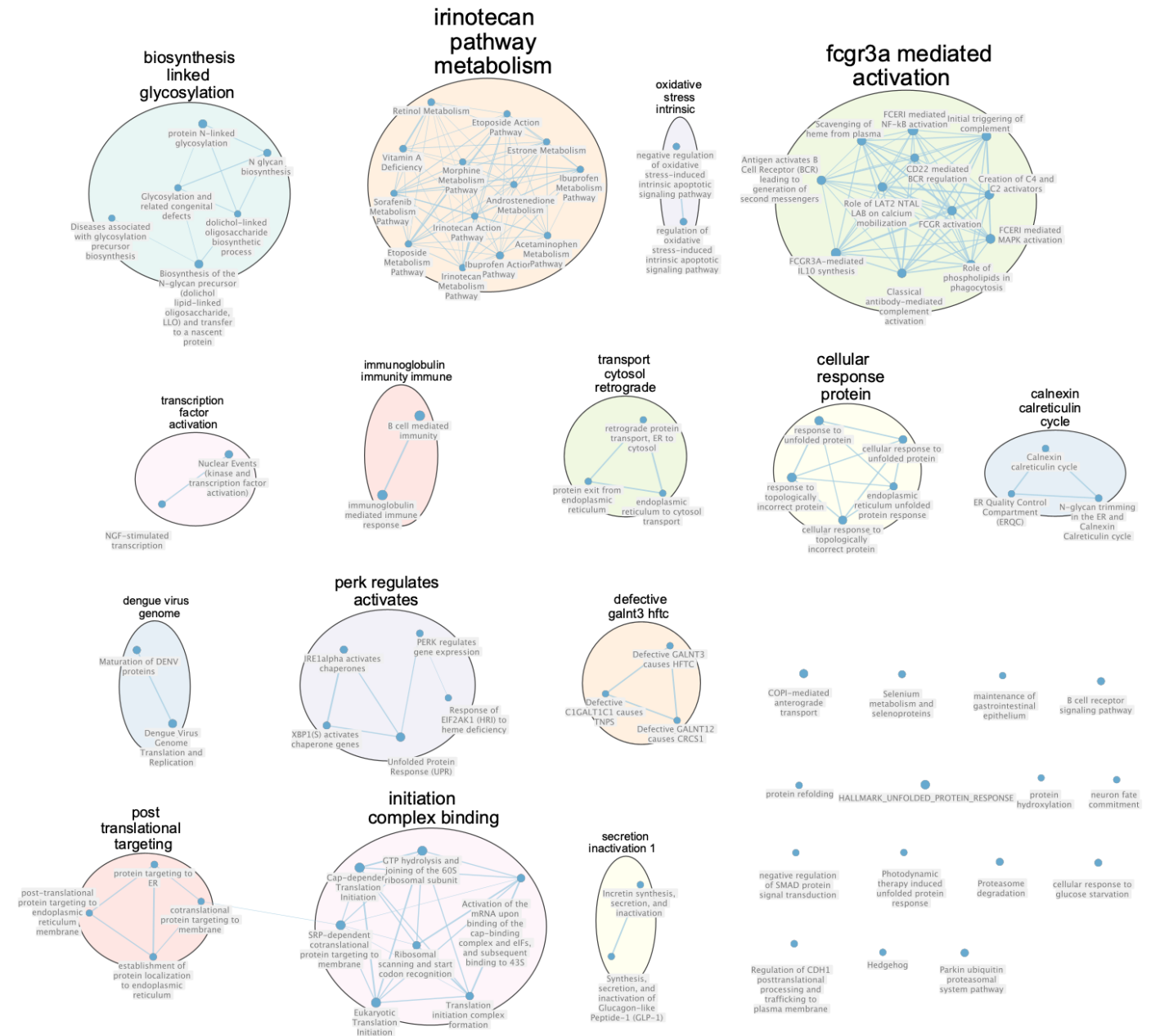
Filter by minimum experiments: no

Parse baderlab names in GMT file: no

Data Set Edges: Automatic

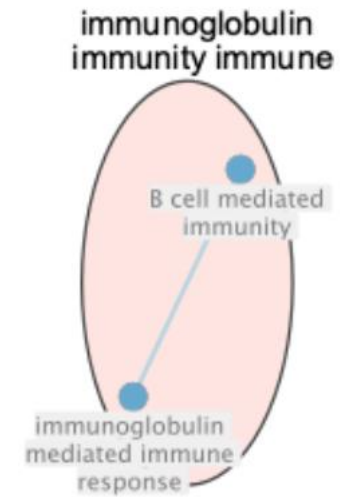
Cutoff: 0.375

Metric: Jaccard + Overlap Combined

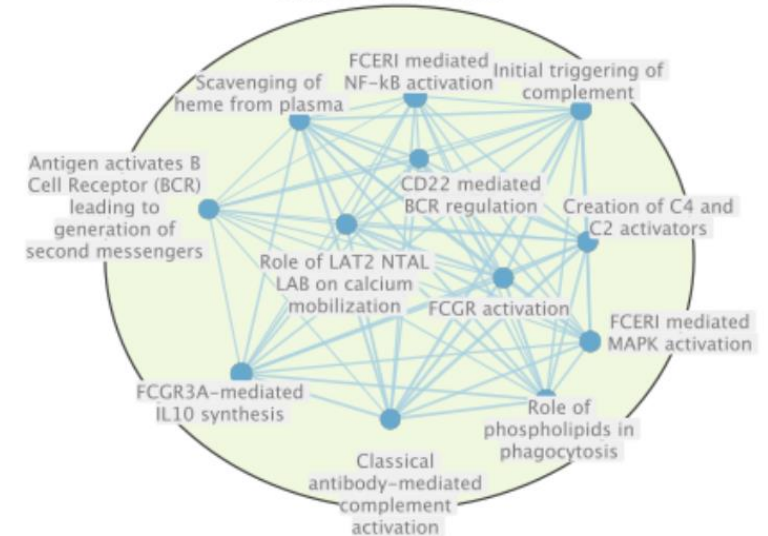


RESULTS – THEMES

1. Immune Signaling and Modulation

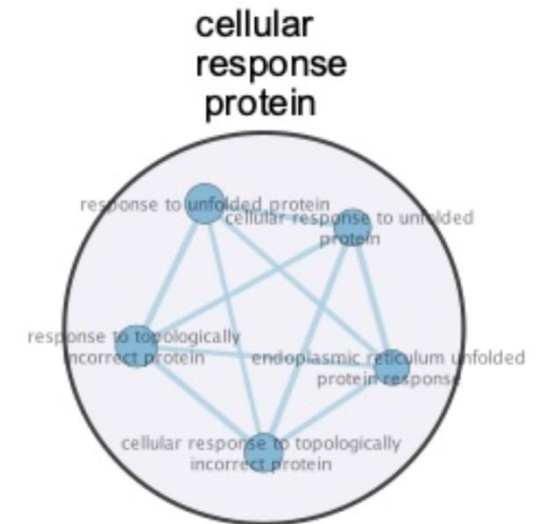
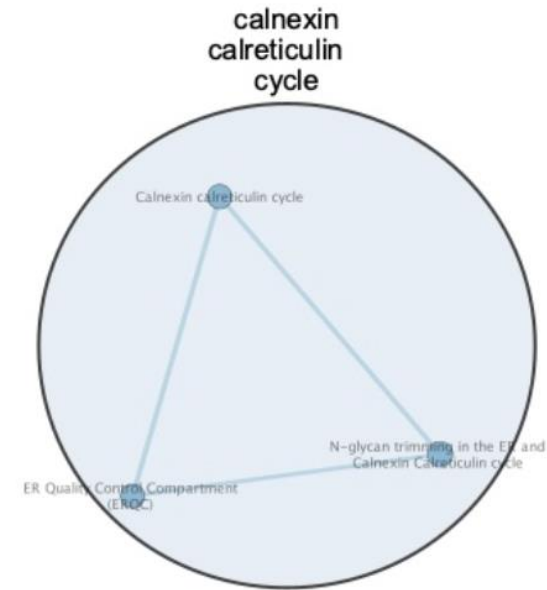


fcgr3a mediated activation



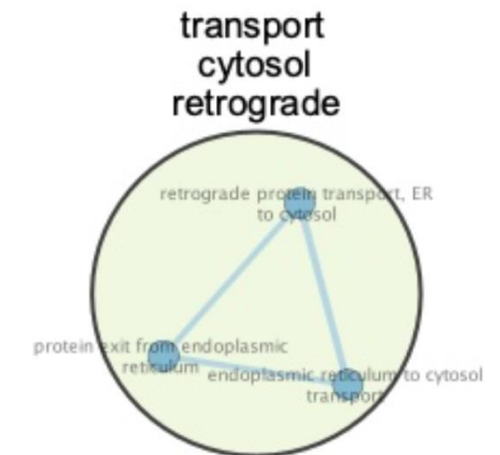
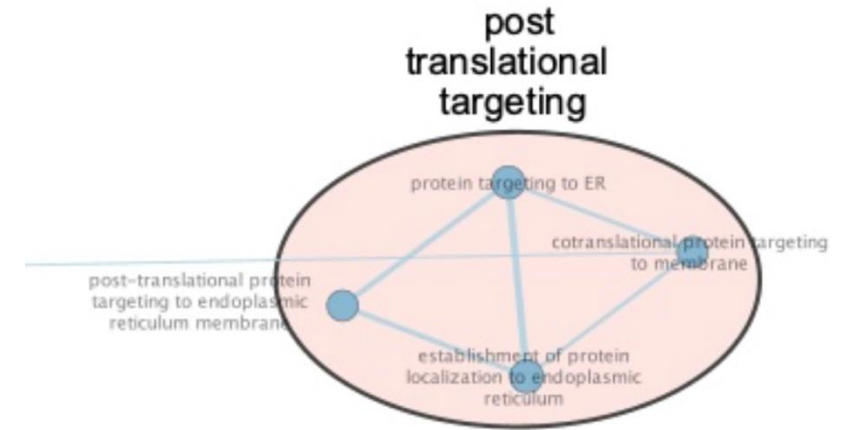
RESULTS – THEMES

1. Immune Signaling and Modulation
2. Protein Folding and Endoplasmic Reticulum (ER) Stress



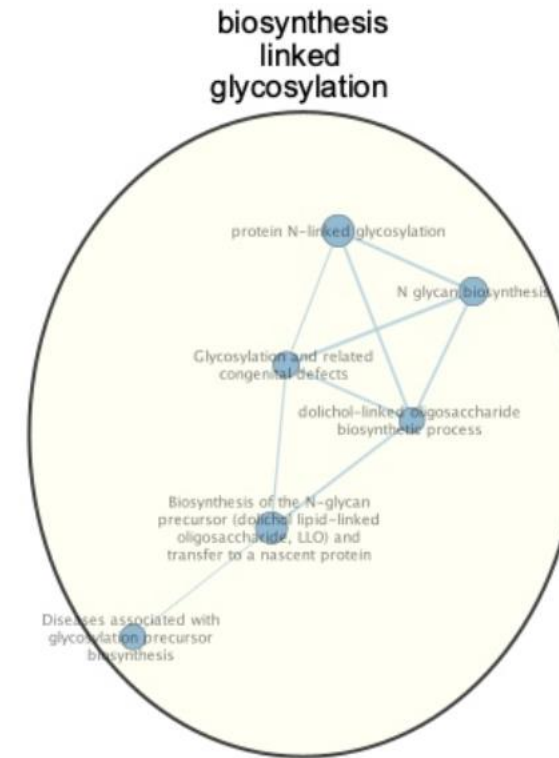
RESULTS – THEMES

1. Immune Signaling and Modulation
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3. Protein Targeting and Translation Initiation



RESULTS – THEMES

1. Immune Signaling and Modulation
2. Protein Folding and Endoplasmic Reticulum (ER) Stress
3. Protein Targeting and Translation Initiation
4. Glycosylation and Barrier Integrity



COMPARISON

Non-threshold enrichment analysis

GSEA identified many more potentially enriched pathways for the same p-value

Threshold enrichment analysis

Enrichment testing on these gene lists returned:

- 16 enriched gene sets for up-regulated genes
- 10 enriched gene sets for down-regulated genes

	na_pos	na_neg
ORA p-value < 1%	16	10
GSEA p-value < 1%	39	108

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GSEA results are significantly more informative regarding the pathophysiology of GvHD

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In the ORA, the results were somewhat fragmented

COMPARISON

Non-threshold enrichment analysis

GSEA identified many more potentially enriched pathways for the same p-value

GSEA results are significantly more informative regarding the pathophysiology of GvHD

GSEA found the functional mechanisms (ZAP-70 translocation, Migration, and Second Messenger generation) that allows those T-cells to cause disease.

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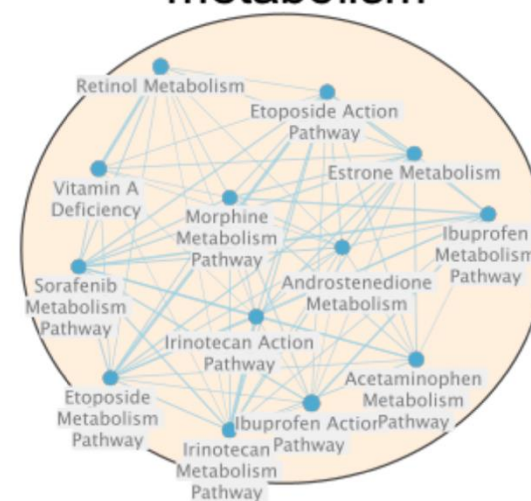
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ORA missing classic “Inflammation” or “T-cell Activation” pathways

KEY INSIGHTS

Novel Pathways: irinotecan pathway

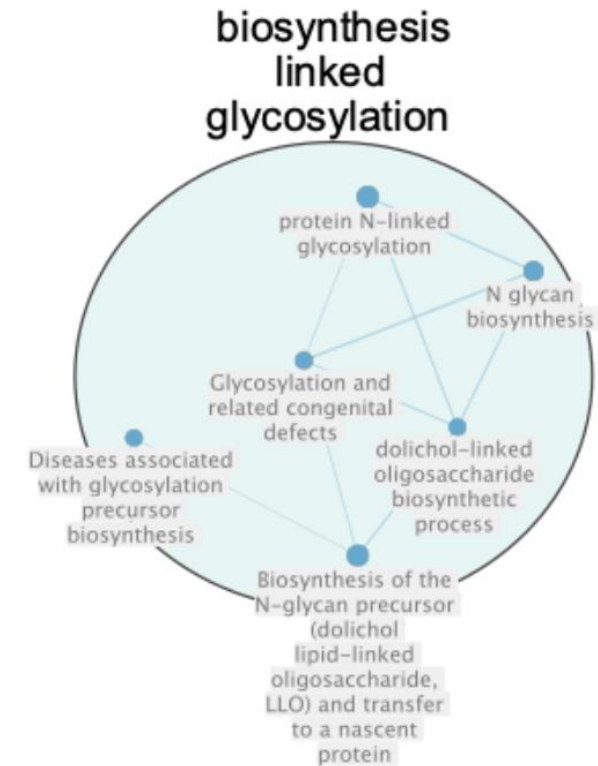
irinotecan
pathway
metabolism



KEY INSIGHTS

Novel Pathways: irinotecan pathway

Support from literature: leaky gut symptoms

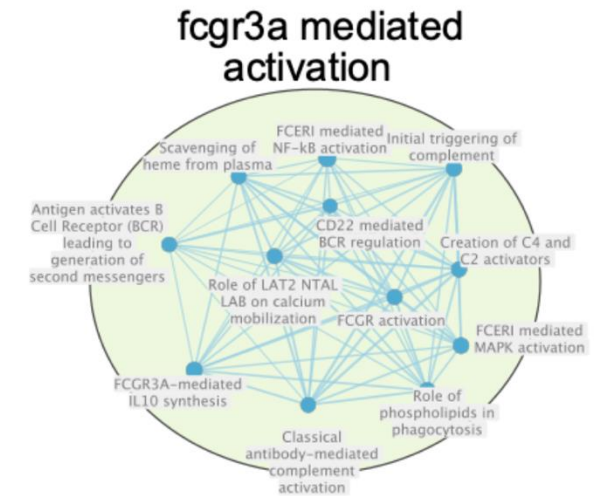
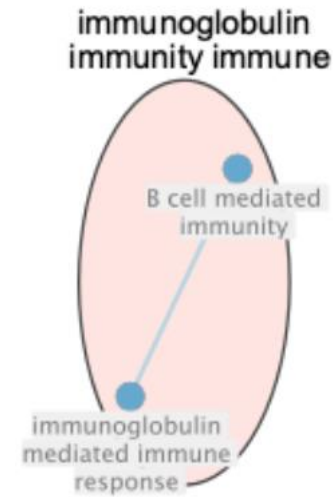


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Novel Pathways: irinotecan pathway

Support from literature: leaky gut symptoms

Support from literature: immune suppression



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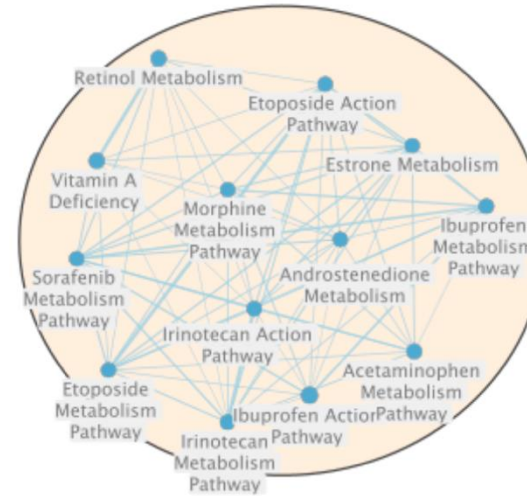
Novel Pathways: irinotecan pathway

Support from literature: leaky gut symptoms

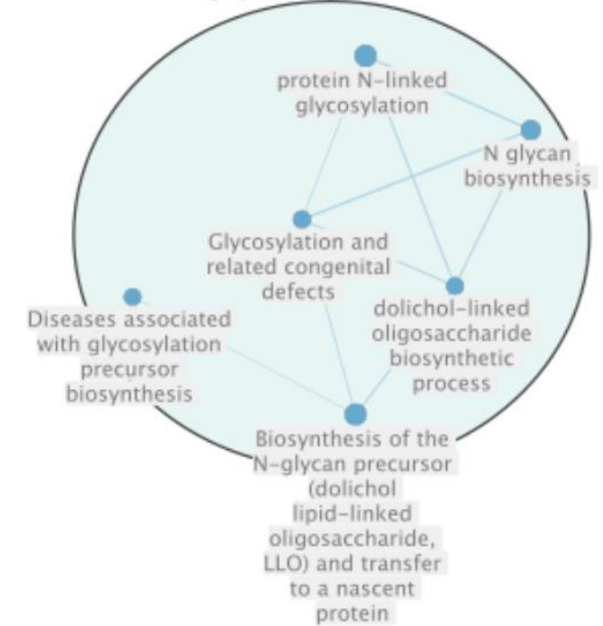
Support from literature: immune suppression

New insights of down regulated pathways

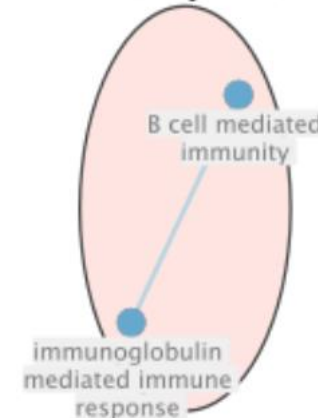
irinotecan pathway metabolism



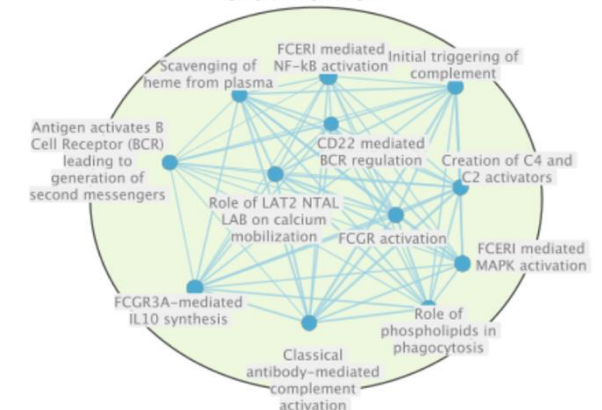
biosynthesis linked glycosylation



immunoglobulin immunity immune



fcgr3a mediated activation



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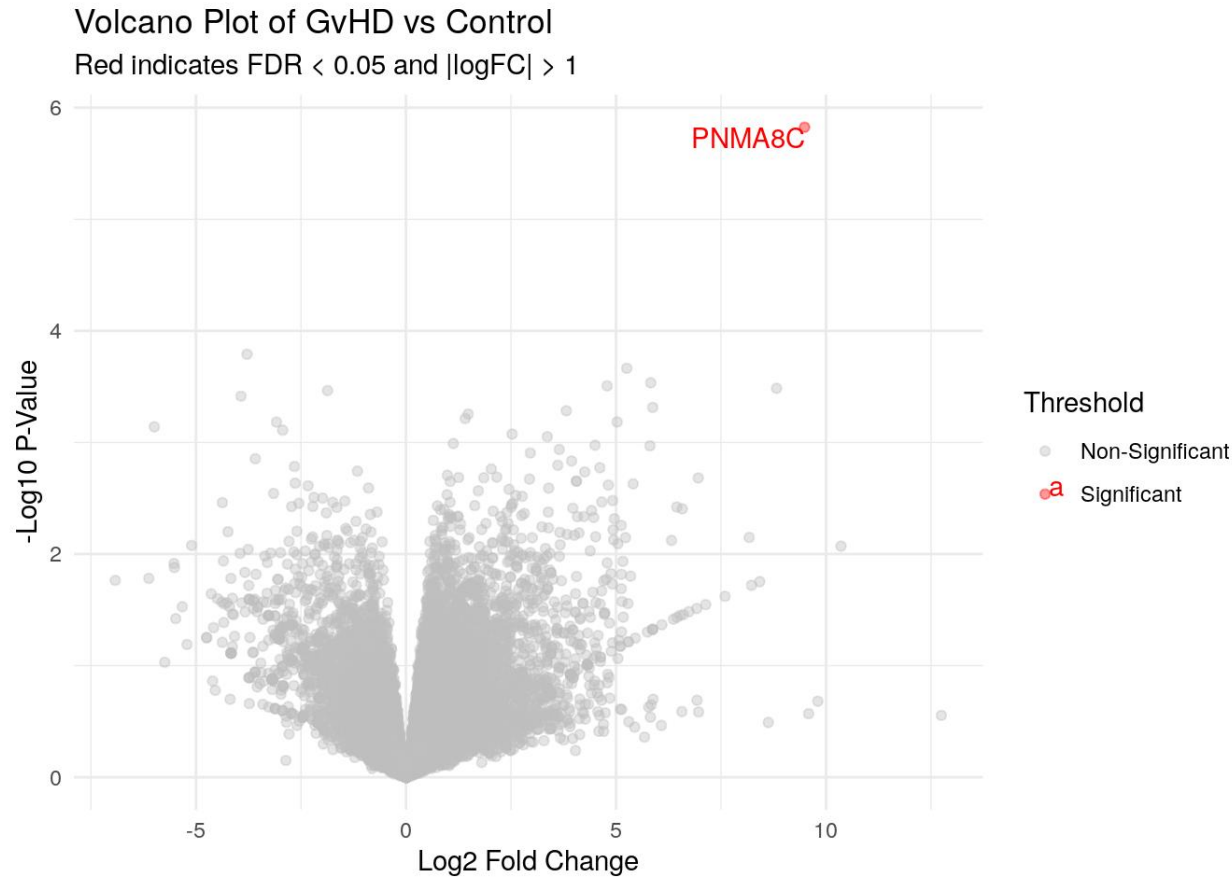
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THANK YOU





(PARANEOPLASTIC MA
ANTIGEN FAMILY MEMBER
8C)

General Function: Members of the PNMA family are often associated with paraneoplastic syndromes (autoimmune responses often triggered by tumors).

Pathway Name

Top 10 Enriched Pathways

